

Firebolt® 8.0 API Specification

Document status	ARCHIVED
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API version	Link	Summary
8.0.1	213	Device.hdr - fixed typo: hd10Plus should be hdr10Plus
8.0.0	180	8.0.0  This is the approved version for RDK8

Key

Color	Meaning
Green	Approved - ready for development
Red	Not approved - not ready for development

Types

	Type	Definition
1	unsigned	32-bit unsigned integer
2	double	64-bit double-precision floating point number

Errors

	Class	Value	Name	Description	Examples	Open issues
1	Generic	-32601	Method not known	The method is not known to the version of Firebolt on this device	Device.foo	
2		-40300	Method not permitted	The app does not have permission to call this method		
3		-32602	Arguments invalid	The set of arguments is not valid		
4		-32100	Unspecified error	Any unexpected error	Memory allocation failure, transport failure	
5	Specific	-50100	Method not supported	A non-mandatory method that is not implemented on this device	SecureStorage.get on a non-Comcast device	
6			App state invalid	The app is not in an appropriate lifecycle state to call this method		

Methods

	Module	Method	Parameters	Returns	Specific errors	API version	C++	JS	Description
1	Accessibility	audioDescription onAudioDescriptionChanged	None	bool	None	8.0.0	✓	✓	Returns the audio description setting of the device
2	Accessibility	closedCaptionsSettings onClosedCaptionsSettingsChanged	None	- enabled - bool - preferredLanguages - list of strings [] (if not initialized) - list of one or more ISO 639-2/B	None	8.0.0	✓	✓	Returns captions settings: enabled, and a list of zero or more languages in order of decreasing preference
3	Accessibility	highContrastUI onHighContrastUIChanged	None	bool	None	8.0.0	✓	✓	Returns the high contrast UI device setting
4	Accessibility	voiceGuidanceSettings onVoiceGuidanceSettingsChanged	None	- enabled - bool - rate - double 1 (normal rate) 0.1 to 10 inclusive - navigationHints - bool	None	8.0.0	✓	✓	Returns voice guidance settings: enabled, rate, and verbosity
5	Actions	start	intent - string	None	None	???	✓	✓	Send an intent to the platform Argument is a JSON document
6	Actions	intent onIntent	None	string	None	???	✓	✓	Returns the intent that was most recently received from the platform, as a JSON document. Getter is typically called by an app after transitioning to the active lifecycle state
7	Advertising	advertisingId	None	- ifa - string - a UUID - ifa_type - string, one of "dpid" - device provided ID "sspid" - SSP provided ID "sessionid" - session /synthetic ID - lmt - string, one of "0" "1"	None	8.0.0	✓	✓	Returns the IFA
8	Device	uid	None	string - a UUID	None	8.0.0	✓	✓	Returns a persistent unique UUID for the current app and device. The UUID is reset when the app or device is reset
9	Device	deviceClass	None	enum - ott - no tuner/demod, no integrated display - stb - with tuner/demod, no integrated display - tv - possibly tuner/demod, with integrated display	None	???	✓	✓	Returns the class of the device
10	Device	uptime	None	unsigned	None	???	✓	✗	Returns the number of seconds since most recent device boot, including any time spent during deep sleep
11	Device	timeInActiveState	None	unsigned	App state invalid	???	✓	✗	Returns the number of seconds since the device transitioned to the ON power state
12	Device	chipsetId	None	string - (see Chipset Id in Devices table)	None	???	✓	✗	Returns chipset ID as a printable string, e.g. "BCM72180"
13	Device	hdr onHdrChanged	None	- hdr10 - bool - hdr10Plus - bool - dolbyVision - bool - hlg - bool	None	8.0.0	✓	✓	Returns the HDR standards that are supported by the attached TV or the integral display - OTT/STB device: supported by the device and indicated by the EDID of the attached TV - TV: supported by the device

14	Discovery	watched	- entityId - string - progress - double - optional - completed - bool - optional - watchedOn - ISO 8601 date and time in UTC - optional - agePolicy - string - optional	None	None	8.0.0	✓	✓	Notify the platform that content has been partially or completely watched progress: VOD: 0 to 0.99, live: number of seconds watchedOn: "YYYY-MM-DDThh:mm:ss.sssZ" agePolicy: app:adult, app:child, app:teen
15	Display	size	None	- width - unsigned - height - unsigned	None	???	✓	✗	Returns the physical dimensions of the connected or integral display, in centimeters Returns 0, 0 on a OTT/STB device when a display is not connected over HDMI Typical values 43" - 97, 56 55" - 123, 71 65" - 145, 83
16	Display	maxResolution	None	- width - unsigned - height - unsigned	None	???	✓	✗	Returns the physical/native resolution of the connected or integral display, in pixels Returns 0, 0 on a OTT/STB device when a display is not connected over HDMI Typical values - HD Ready - 1280, 720 - Full HD - 1920, 1080 - UHD - 3840, 2160
17	Display	edid	None	string - Base64	None	???	✓	✗	Returns the EDID (and extensions) of the connected or integral display, as a Base64 encoded string Returns an empty string, "", on a OTT/STB device when a display is not connected over HDMI
18	Firebolt	apiVersion	None	- version - string - major - unsigned - minor - unsigned - patch - unsigned	None	???	✓	✓	Returns the Firebolt API version of the client library as a semantic versioning string, and unsigned numbers representing the same version Note: the version returned is of the client library, not the platform
19	Lifecycle2	close	type - enum	- deactivate - unload - killReload - killReactivate	None	8.0.0	✓	✗	Request the platform to deactivate the app, and possibly take further action type - deactivate - app is deactivated (if active) - typically a user closing an app - unload - app is deactivated (if active) and terminated - typically an app has detected an error - killReload - app is killed, then reloaded to paused or suspended (by policy) - killReactivate - app is killed, reloaded, then activated window.close() maps to unload window.minimize() maps to deactivate Module is named Lifecycle in C++ Client Library
20	Lifecycle2	state	None	enum	- initializing - active - paused - suspended - hibernated - terminating	8.0.0	✓	✗	Returns the current lifecycle state of the app. The first lifecycle state that can be observed by an app/runtime is the initializing state Apps/runtimes should ordinarily use state change notifications rather than this method

21	Lifecycle2	onStateChanged	List, of length one, of the lifecycle state change - oldState - enum - newState - enum	N/A	N/A	8.0.0	✓	✗	Subscribe to/unsubscribe from lifecycle state changes. The app/runtime remains in the initializing state until the subscribe call is made to the platform Notification of lifecycle state change, raised after the platform has transitioned the app/runtime to the new lifecycle state Note: the list always carries exactly one lifecycle state change Valid transitions - initializing to paused - initializing to suspended - paused to active - active to paused - paused to suspended - suspended to paused - suspended to hibernated - hibernated to suspended - any to terminating
22	Localization	country onCountryChanged	None	string, either "" (if not initialized) - ISO 3166-1 alpha-2 (see Country in Devices table)	None	8.0.0	✓	✓	Return the country, e.g. US, CA, GB, BE
23	Localization	preferredAudioLanguages onPreferredAudioLanguagesChanged	None	list of strings, either [] (if not initialized) - list of one or more ISO 639-2/B (see Secondary Audio Language in Devices table)	None	8.0.0	✓	✓	A list of zero or more languages in order of decreasing preference. Typically two languages are present. A language may be repeated in the list
24	Localization	presentationLanguage onPresentationLanguageChanged	None	string, either "" (if not initialized) - BCP 47 (see Presentation Languages in Devices table)	None	8.0.0	✓	✓	The presentation language of the device, e.g. en-US
25	Metrics	ready	None	None	None	8.0.0	✓	✓	Inform the platform that the app is minimally usable. This method is called automatically by Lifecycle.ready()
26	Metrics	signIn	None	None	None	8.0.0	✓	✗	Log a sign in event, called by Discovery. signIn()
27	Metrics	signOut	None	None	None	8.0.0	✓	✗	Log a sign out event, called by Discovery. signOut()
28	Metrics	startContent	- entityId - string - optional - agePolicy - string - optional	None	None	8.0.0	✓	✓	Inform the platform that your user has started content
29	Metrics	stopContent	- entityId - string - optional - agePolicy - string - optional	None	None	8.0.0	✓	✓	Inform the platform that your user has stopped content
30	Metrics	page	- pageId - string - agePolicy - string - optional	None	None	8.0.0	✓	✓	Inform the platform that your user has navigated to a page or view
31	Metrics	error	- type - enum - network - media - restriction - entitlement - other - code - string - description - string - visible - bool - parameters - arg list - optional - agePolicy - string - optional	None	None	8.0.0	✓	✓	Inform the platform of an error that has occurred in your app

32	Metrics	mediaLoadStart	- entityId - string - agePolicy - string - optional	None	None	8.0.0	✓	✓	Called when setting the URL of a media asset to play, in order to infer load time
33	Metrics	mediaPlay	- entityId - string - agePolicy - string - optional	None	None	8.0.0	✓	✓	Called when media playback should start due to autoplay, user-initiated play, or unpausing
34	Metrics	mediaPlaying	- entityId - string - agePolicy - string - optional	None	None	8.0.0	✓	✓	Called when media playback actually starts due to autoplay, user-initiated play, unpausing, or recovering from a buffering interruption
35	Metrics	mediaPause	- entityId - string - agePolicy - string - optional	None	None	8.0.0	✓	✓	Called when media playback will pause due to an intentional pause operation
36	Metrics	mediaWaiting	- entityId - string - agePolicy - string - optional	None	None	8.0.0	✓	✓	Called when media playback will halt due to a network, buffer, or other unintentional constraint
37	Metrics	mediaSeeking	- entityId - string - target - double - agePolicy - string - optional	None	None	8.0.0	✓	✓	Called when a seek is initiated during media playback
38	Metrics	mediaSeeked	- entityId - string - position - double - agePolicy - string - optional	None	None	8.0.0	✓	✓	Called when a seek is completed during media playback
39	Metrics	mediaRateChanged	- entityId - string - rate - double - agePolicy - string - optional	None	None	8.0.0	✓	✓	Called when the playback rate of media is changed
40	Metrics	mediaRenditionChanged	- entityId - string - bitrate - unsigned - width - unsigned - height - unsigned - profile - string - optional - agePolicy - string - optional	None	None	8.0.0	✓	✓	Called when the playback rendition (e.g. bitrate, dimensions, profile, etc) is changed
41	Metrics	mediaEnded	- entityId - string - agePolicy - string - optional	None	None	8.0.0	✓	✓	Called when playback has stopped because the end of the media was reached
42	Metrics	event	- schema - uri - data - string - agePolicy - string - optional	None	None	8.0.0	✓	✓	Inform the platform of 1st party distributor metrics data parameter is a JSON document
43	Metrics	appInfo	build - string	None	None	8.0.0	✓	✓	Inform the platform about an app's build info
44	Network	connected onConnected Changed	None	bool	None	8.0.0	✓	✓	Returns whether the device has a useable network connection
45	ParentalControl	pinControl	None	bool	-50100 / Not supported	???	✓	✓	Returns whether PIN blocking is enabled, or an error if this setting is not exposed to apps
46	ParentalControl	blockNotRatedContent	None	bool	-50100 / Not supported	???	✓	✓	Returns whether content that is not rated should be blocked, or an error if this setting is not exposed to apps
47	ParentalControl	viewingRestrictions	None	string	-50100 / Not supported	???	✓	✓	Returns a JSON document describing the ratings schemes configured for the device and any ratings that are blocked for that scheme, or an error if this setting is not exposed to apps

48	Presentation	focused onFocusedC hanged	None	bool	None	8.0.0	✓	✗	Whether the app is in focus, i.e. receiving key presses. Provided for those apps/runtimes that cannot use Wayland
49	Stats	memoryUsage	None	• userMemoryUsed - unsigned • userMemoryLimit - unsigned • gpuMemoryUsed - unsigned • gpuMemoryLimit - unsigned	None	???	✓	✗	Returns information about container memory usage, in bytes
50	TextToSpeech	speak	text - string	• speechid - unsigned • TTS_Status - 0...3 • success - bool	None	8.0.0	✓	✗	Speak the utterance immediately. Any ongoing speech is interrupted Text argument is either plain text or a well-formed SSML document TTS_Status, not success attribute, to be used by caller to indicate success of call 0 OK, 1 Fail, 2 not enabled, 3 invalid configuration Raises onSpeechinterrupted if speaking is interrupted
51	TextToSpeech	pause	speechid - unsigned	• TTS_Status - 0...3 • success - bool	None	8.0.0	✓	✗	Pauses the utterance Raises onSpeechpause if ongoing speech is paused Does nothing if utterance is already paused
52	TextToSpeech	resume	speechid - unsigned	• TTS_Status - 0...3 • success - bool	None	8.0.0	✓	✗	Continue the paused utterance Raises onSpeechresume if paused speech is resumed Does nothing if the utterance is not paused
53	TextToSpeech	cancel	speechid - unsigned	• TTS_Status - 0...3 • success - bool	None	8.0.0	✓	✗	Stop speaking if utterance is currently being spoken Raises onSpeechinterrupted if speaking was interrupted
54	TextToSpeech	getspeechstate	speechid - unsigned	• speechstate - enum ◦ SPEECH_PENDING ◦ SPEECH_IN_PROGRESS ◦ SPEECH_PAUSED ◦ SPEECH_NOT_FOUND • TTS_Status - 0...3 • success - bool	None	8.0.0	✓	✗	Returns the state of the utterance
55	TextToSpeech	onWillSpeak	speechid - unsigned	N/A	N/A	8.0.0	✓	✗	Text to speech conversion is about to start
56	TextToSpeech	onSpeechstart	speechid - unsigned	N/A	N/A	8.0.0	✓	✗	Utterance is about to be spoken
57	TextToSpeech	onSpeechpause	speechid - unsigned	N/A	N/A	8.0.0	✓	✗	Ongoing speech was paused
58	TextToSpeech	onSpeechresume	speechid - unsigned	N/A	N/A	8.0.0	✓	✗	Paused speech was resumed
59	TextToSpeech	onSpeechcomplete	speechid - unsigned	N/A	N/A	8.0.0	✓	✗	Speech completed successfully
60	TextToSpeech	onSpeechinterrupted	speechid - unsigned	N/A	N/A	8.0.0	✓	✗	Speech was stopped, due to another call to speak or cancel
61	TextToSpeech	onNetworkerror	speechid - unsigned	N/A	N/A	8.0.0	✓	✗	Utterance failed due to network
62	TextToSpeech	onPlaybackerror	speechid - unsigned	N/A	N/A	8.0.0	✓	✗	Utterance failed during playback
63	TextToSpeech	listvoices	language - string - BCP 47	• TTS_Status - 0...3 • voices - list of one or more strings	None	8.0.0	✓	✗	Returns the list of available voices as human-readable strings, e.g. "ava", "amelie", "angelica"